

Schrödinger Bridge for Non-linear ROMs

Reference Drifts, FFD Registration and the γ -Sweep

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Recap — Fokker–Planck Formulation

Let $\mu, \nu \in \mathcal{P}(\mathbb{R}^d)$ be the initial and terminal marginals, time rescaled to $[0, 1]$, $\gamma > 0$ the diffusion coefficient.

We seek a density–velocity pair $(\rho_t, \mathbf{b}_t)_{t \in [0, 1]}$ with $\rho_0 = \mu$, $\rho_1 = \nu$ such that

- ρ_t solves the **Fokker–Planck equation**

$$\partial_t \rho_t + \nabla \cdot (\rho_t \mathbf{b}_t) = \gamma \Delta \rho_t,$$

- the pair minimizes the **expected kinetic energy**

$$\mathcal{J}(\rho, \mathbf{b}) = \int_0^1 \int_{\mathbb{R}^d} \frac{1}{2} |\mathbf{b}_t(\mathbf{x})|^2 \rho_t(\mathbf{x}) \, d\mathbf{x} \, dt.$$

Girsanov applied to $dX_t = \mathbf{b}_t(X_t) \, dt + \sqrt{2\gamma} \, dW_t$ gives $\text{KL}(\mathbb{P} \parallel \mathbb{R}) = \mathcal{J}/(2\gamma)$: the Schrödinger bridge is the minimal-energy stochastic interpolation between μ and ν .

Recap — Heat and Advection–Diffusion Semigroups

Hopf–Cole ($\eta = e^f$, $\eta^* = e^g$) linearizes the optimality system into two transport–diffusion PDEs, whose transition operators are the SB kernel.

Driftless bridge ($\beta \equiv 0$) — heat semigroup $\mathcal{P}_\tau = e^{\tau\gamma\Delta}$

$$\partial_\tau \phi = \gamma \Delta \phi, \quad \partial_n \phi|_{\partial\Omega} = 0 \quad (\text{Neumann, no-flux}).$$

Drifted bridge — advection–diffusion semigroups $\mathcal{Q}_\tau = e^{\tau\mathcal{L}}$, $\mathcal{Q}_\tau^\dagger = e^{\tau\mathcal{L}^*}$

$$\underbrace{\partial_\tau \phi = \beta \cdot \nabla \phi + \gamma \Delta \phi}_{\mathcal{Q}: \text{non-conservative, backward}}, \quad \underbrace{\partial_\tau \phi = -\nabla \cdot (\beta \phi) + \gamma \Delta \phi}_{\mathcal{Q}^\dagger: \text{conservative, forward}}.$$

IPFP in log-potential form:

$$f^{(k+1)} = \log \mu - \log \mathcal{Q}_1[e^{g^{(k)}}], \quad g^{(k+1)} = \log \nu - \log \mathcal{Q}_1^\dagger[e^{f^{(k+1)}}].$$

FVM: TPFA diffusion + first-order upwind advection, SSP-RK2, Anderson-accelerated.

$$\beta \equiv 0 \Rightarrow \mathcal{Q}_1 = \mathcal{Q}_1^\dagger = \mathcal{P}_1.$$

Recap — Convex Displacement Interpolation (CDI)

The maps act on the shock-feature marginals; the physical Mach field is recovered gradient-free. For $t \in (0, 1)$,

$$\hat{M}(t, \mathbf{x}) = (1 - t) M_0((1 - t) \mathbf{x} + t S(\mathbf{x})) + t M_1(t \mathbf{x} + (1 - t) T(\mathbf{x}))$$

- M_0, M_1 = endpoint Mach fields, sampled at the displaced pre-images by linear interpolation on the mesh;
- exact at the endpoints: $\hat{M}(0, \cdot) = M_0$, $\hat{M}(1, \cdot) = M_1$;
- no map inversion — forward pre-image $t\mathbf{x} + (1 - t)T$ pulls back M_1 , backward pre-image $(1 - t)\mathbf{x} + tS$ pulls back M_0 , blended with weights $(1 - t, t)$.

The reference drift is designed to keep T, S close to the identity, so the CDI reconstruction stays well-conditioned at small γ .

Recap — Construction of the Transport Maps T, S

T is the conditional mean of the target position given the source (built from g through Q_1); S is its backward analogue (from f through $Q_1^\dagger$):

$$T(\mathbf{x}) = \frac{Q_1[\text{id } e^g](\mathbf{x})}{Q_1[e^g](\mathbf{x})} - \beta_{\text{bias}}(\mathbf{x}), \quad S(\mathbf{x}) = \frac{Q_1^\dagger[\text{id } e^f](\mathbf{x})}{Q_1^\dagger[e^f](\mathbf{x})} - \beta_{\text{bias}}(\mathbf{x}).$$

Neumann identity-leak correction

$$\beta_{\text{bias}} = \frac{Q_1[\text{id}]}{Q_1[1]} - \Phi_{\mathbf{b}},$$

measured against the drift's own *reflected deterministic flow* $\Phi_{\mathbf{b}}$ (identity when $\mathbf{b} \equiv 0$), so $\mathbf{b}$'s transport stays in T, S .

Evaluated in the stable log-domain with a positivity shift $\text{id} \mapsto \text{id} - c$; a trust gate reverts kernel-collapse outlier cells to the identity. $T, S \rightarrow \text{id}$ far from the transported feature.

Oblique Drift (1/2) — Wave Angles

The LE and TE shocks rotate between their M_0 and M_1 wave angles:

θ – β – M relation (weak root) — LE family

$$\tan \theta = 2 \cot \beta_s \frac{M^2 \sin^2 \beta_s - 1}{M^2(\gamma_g + \cos 2\beta_s) + 2}, \quad \beta_s^{\text{LE}}(M) = \beta_s(\alpha, M).$$

TE family — LE shock $\rightarrow$ Prandtl–Meyer(2α) $\rightarrow$ TE shock

$$\nu(M_b) = \nu(M_a) + 2\alpha, \quad \beta_s^{\text{TE}}(M) = \beta_s(\alpha, M_b(M)).$$

Ray angles and rotation rates, $M(t) = (1 - t)M_0 + tM_1$

$$\psi_{\text{LE}} = \beta_s^{\text{LE}}, \quad \psi_{\text{TE}} = \beta_s^{\text{TE}} - \alpha, \quad \omega_\kappa(t) = \left. \frac{d\beta_s^\kappa}{dM} \right|_{M(t)} (M_1 - M_0).$$

Oblique Drift (2/2) — Windowed Rigid Rotation

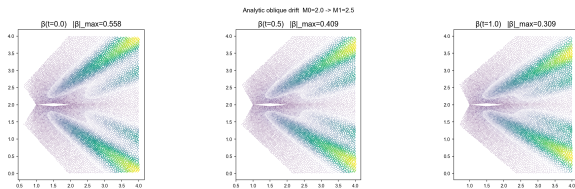
Four windowed rigid rotations about the apexes $\mathbf{a}_\kappa$

$$\boldsymbol{\beta}(t, \mathbf{x}) = \sum_{\kappa \in \{\text{LE, TE}\}} \sum_{\sigma = \pm 1} \sigma \omega_\kappa(t) w_{\kappa, \sigma}(t, \mathbf{x}) \mathbf{R}(\mathbf{x} - \mathbf{a}_\kappa), \quad \mathbf{R} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

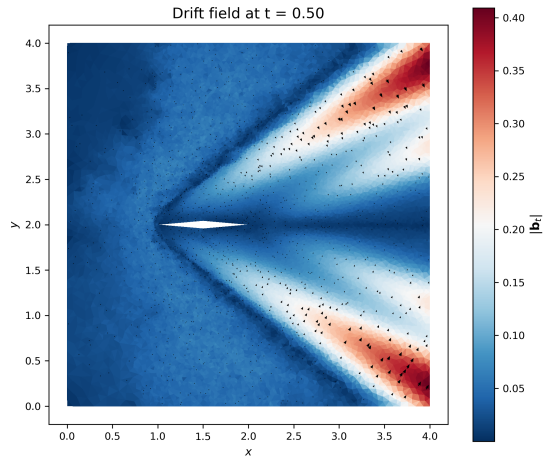
$$w_{\kappa, \sigma} = e^{-\text{dist}_{\kappa, \sigma}^2 / \sigma_w(s)^2} \mathbf{1}[s_{\kappa, \sigma} > 0], \quad s_{\kappa, \sigma} = (\mathbf{x} - \mathbf{a}_\kappa) \cdot \hat{\mathbf{d}}_{\kappa, \sigma}.$$

Zero in the freestream; the conical band $\sigma_w(s)$ covers the swept wedge; $\nabla \cdot \boldsymbol{\beta} \neq 0 \Rightarrow$ enters through the conservative $Q^\dagger$.

Oblique Drift — Field



Analytic $\beta(t=0.5)$



$$\text{Total drift } \mathbf{b}_t = \beta + 2\gamma \nabla g_t$$

Free-Form Deformation (FFD)

$$T : \mathbb{R}^{N_{cp}} \times \mathbb{R}^3 \longrightarrow \mathbb{R}^3$$

$$\lambda, \xi \longmapsto T_{\lambda_0}(\xi) = x$$

In the coordinate system $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$

$$\mathbf{x} = \mathbf{x}_0 + t_1 \mathbf{e}_1 + t_2 \mathbf{e}_2 + t_3 \mathbf{e}_3$$

The mapping is defined as:

$$FFD(t_1, t_2, t_3) = \sum_{i=0}^l \sum_{j=0}^m \sum_{k=0}^n B_i^l(t_1) B_j^m(t_2) B_k^n(t_3) P_{i,j,k} \quad (1)$$

Here, $P_{i,j,k}$ are the control points in the deformed lattice, and l, m, n denote the degree of the polynomial in each direction.

The (t_1, t_2, t_3) coordinates can be calculated using:

$$t_1 = \frac{(\mathbf{e}_2 \times \mathbf{e}_3) \cdot (X - X_0)}{(\mathbf{e}_2 \times \mathbf{e}_3) \cdot \mathbf{e}_1} \quad (2)$$

$$t_2 = \frac{(\mathbf{e}_1 \times \mathbf{e}_3) \cdot (X - X_0)}{(\mathbf{e}_1 \times \mathbf{e}_3) \cdot \mathbf{e}_2} \quad (3)$$

$$t_3 = \frac{(\mathbf{e}_1 \times \mathbf{e}_2) \cdot (X - X_0)}{(\mathbf{e}_1 \times \mathbf{e}_2) \cdot \mathbf{e}_3} \quad (4)$$

Where a Bernstein polynomial is defined as:

$$B_i^l(t) = \binom{l}{i} t^i (1-t)^{l-i} \quad (5)$$

Data-Driven Drift — FFD Registration

Register the two shock-feature marginals u_0, u_1 by two FFD maps T (forward) and W (backward), whose control-point displacements $\Delta P^T, \Delta P^W$ are the free DOF (single region, all control points free, no boundary conditions).

Objective (single region), weights $(\alpha_A, \alpha_B, \alpha_C, \alpha_D, \alpha_G) = (1, 0.1, 1, 10^{-2}, 0.1)$

$$\begin{aligned} J = & \alpha_A (\|u_1 \circ T - u_0\|^2 + \|u_0 \circ W - u_1\|^2) \\ & + \alpha_B (\|W \circ T - \text{id}\|^2 + \|T \circ W - \text{id}\|^2) \\ & + \alpha_C (\|b_\varepsilon(\det \nabla T)\|^2 + \|b_\varepsilon(\det \nabla W)\|^2) + \frac{\alpha_D}{N_{cp}} (\|\Delta P^T\|^2 + \|\Delta P^W\|^2) \\ & + \alpha_G (\mathcal{W}_2^2(u_1 \circ T, u_0) + \mathcal{W}_2^2(u_0 \circ W, u_1)) \end{aligned}$$

α_A alignment (both directions) · α_B bijectivity / map-sync · α_C Jacobian log-barrier (positivity) · α_D control-point reg. · α_G $\mathcal{W}_2$ capture.

T, W analytic (Bernstein) $\Rightarrow$ evaluated *exactly* at the true cell centres — no grid $\rightarrow$ mesh interpolation error.

From FFD Mappings to the Reference Drift β

The alignment term drives $u_1(T(\mathbf{x})) = u_0(\mathbf{x})$: at a field-0 feature p_0 , $T(p_0) = p_1$ (its field-1 position). Evaluating the converged Bernstein maps at the true cell centres and taking the frozen unit-scale ($s=1$) displacements,

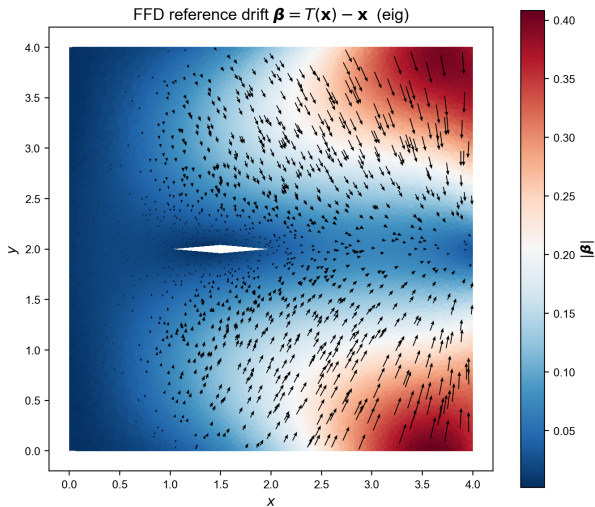
$$\beta(\mathbf{x}) = T(\mathbf{x}, s=1) - \mathbf{x}$$

$$\alpha(\mathbf{x}) = W(\mathbf{x}, s=1) - \mathbf{x}$$

forward ($\mu_0 \rightarrow \mu_1$) via β ; backward ($W \approx T^{-1}$) via α .

- β is a frozen $(N, 2)$ field, consumed by the SB kernel $\mathcal{Q}/\mathcal{Q}^\dagger$, $\Phi_{\mathbf{b}}$ -referenced bias, trust gate and CDI *exactly* like the oblique drift;
- β is cached per marginal pair (`ffd_beta_⟨hash⟩.npz`) — the registration is offline, run once;
- the raw maps $T = \mathbf{x} + \beta$, $S = \mathbf{x} + \alpha$ also serve as a *stand-alone* interpolator (FFD control).

FFD Reference Drift $\beta = T(\mathbf{x}) - \mathbf{x}$



Take the reference drift from the *previous* γ -stage's own solution, carrying its full time dependence:

$$\gamma_0 > \gamma_1 > \cdots > \gamma_K, \quad \beta^{(k+1)}(t) = \beta^{(k)}(t) + 2\gamma_k \nabla g_t^{(k)},$$

stored as a $K \times N \times 2$ stack and fed to the kernel at every substep.

Why it is well posed

β optimal $\Rightarrow \nabla g \rightarrow 0 \Rightarrow \beta' = \beta$: the SB solution is a fixed point of the map.

Time direction

$\mathcal{Q}$ reads β *backward* $(1 - \tau)$, $\mathcal{Q}^\dagger$ *forward* (τ) .
Mirrored: $\langle \mathcal{Q}u, v \rangle = \langle u, \mathcal{Q}^\dagger v \rangle$ to 7×10^{-15} .

SB² — Conditioning: FFD Seed and Endpoint Clip

1. Seed and switch

$\beta^{(0)} = \beta_{\text{FFD}}$, held **fixed** while $\gamma > \gamma_\star = 10^{-3}$; self-conditioning below.

A heat-seeded start recovers a large, void-dominated $2\gamma\nabla g$ and loses the whole mid-ladder.

2. Endpoint clip

$g_t = \log \mathcal{Q}_{1-t}[e^g]$ and $\mathcal{Q}_0 = \text{Id}$, so at $t = 1$ the *raw* log-potential's gradient enters β .

Sample $t \in [c, 1 - c]$ with $c = 0.2$.

t	0.00	0.25	0.50	0.75	0.90	1.00
$\max 2\gamma\nabla g_t $	1.35	1.50	1.79	2.40	3.70	13.76

$|\beta|_{\max}$ sets the advective CFL for every stage, so one singular slice slows the whole ladder.

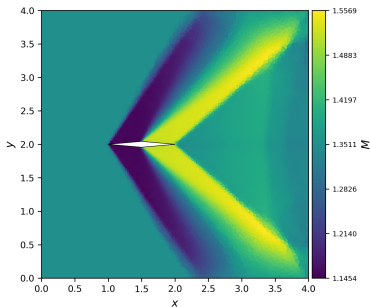
Mach bands	0.80–0.90, ..., 2.90–3.00 (22, width 0.10)
AoA bands	0–1, 1–2, 2–3, 3–4 deg (4) at $M = 2.00$
Incidence	AoA = 0°, 2° for the Mach axis
Drifts	null, oblique, ffd, SBsquared_exact+ffd
$\gamma_{\min}$	10^{-2} , 10^{-3} , 10^{-4} (ladder prefixes)
Mesh	$h = 0.025$ (20 190 cells); $h = 0.0175$, 0.0125 for the h -study
Runs	1158 diamond SB runs; 481 HR Euler snapshots

Each band needs its 2 endpoints + 9 interior references at $t = k/10$, hence a 0.01 Mach grid.

Snapshots — $M : 1.30 \rightarrow 1.40$ (low supersonic), $t = 0.5$

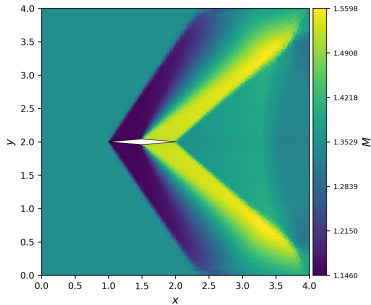
BaryCDI

SB-CDI (barycentric) $t=0.50$



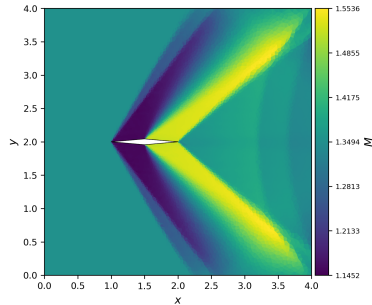
FFD

FFD registration $t=0.50$

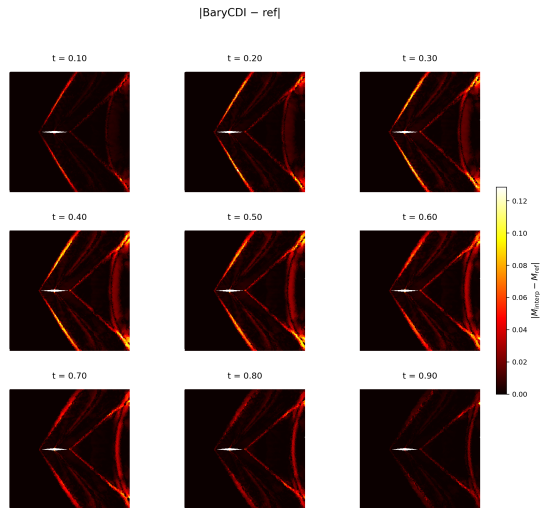


Linear

Linear $t=0.50$



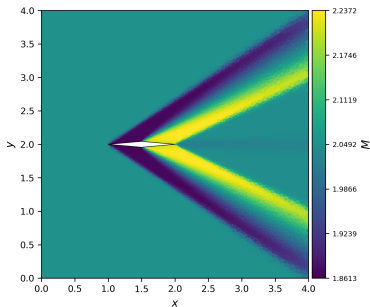
Absolute Error — $M : 1.30 \rightarrow 1.40$ (low supersonic)



Snapshots — $M : 2.00 \rightarrow 2.10$ (mid), $t = 0.5$

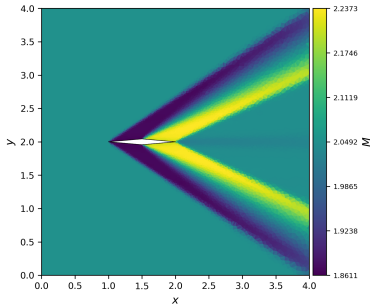
BaryCDI

SB-CDI (barycentric) $t=0.50$



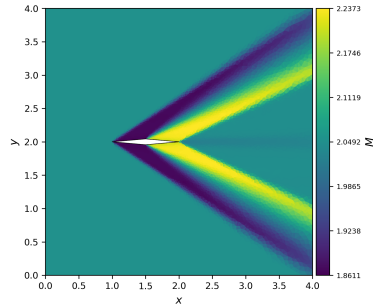
FFD

FFD registration $t=0.50$

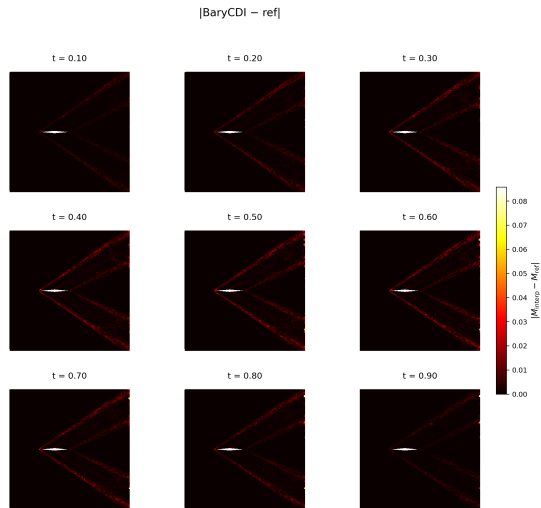


Linear

Linear $t=0.50$



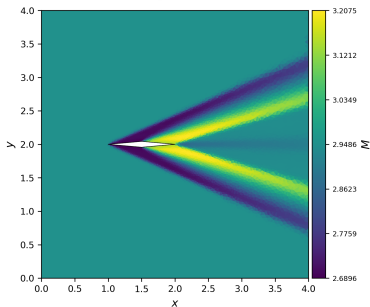
Absolute Error — $M : 2.00 \rightarrow 2.10$ (mid)



Snapshots — $M : 2.90 \rightarrow 3.00$ (high), $t = 0.5$

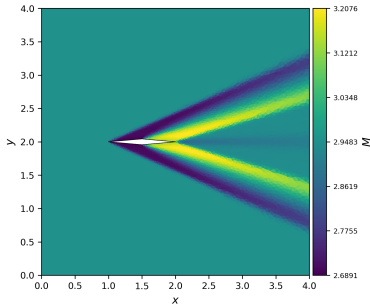
BaryCDI

SB-CDI (barycentric) $t=0.50$



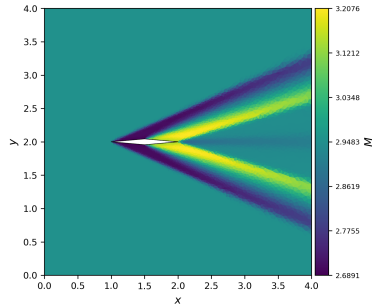
FFD

FFD registration $t=0.50$

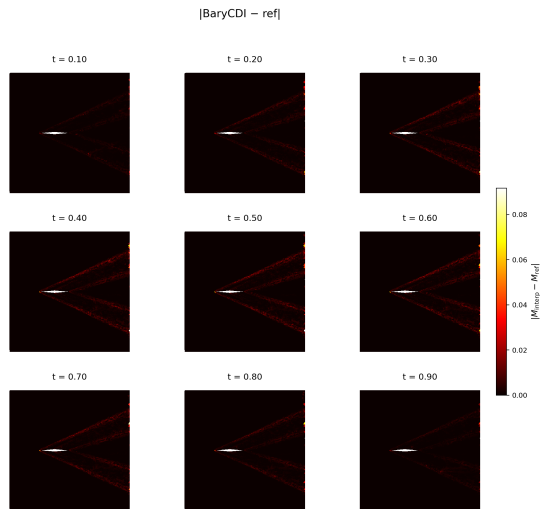


Linear

Linear $t=0.50$

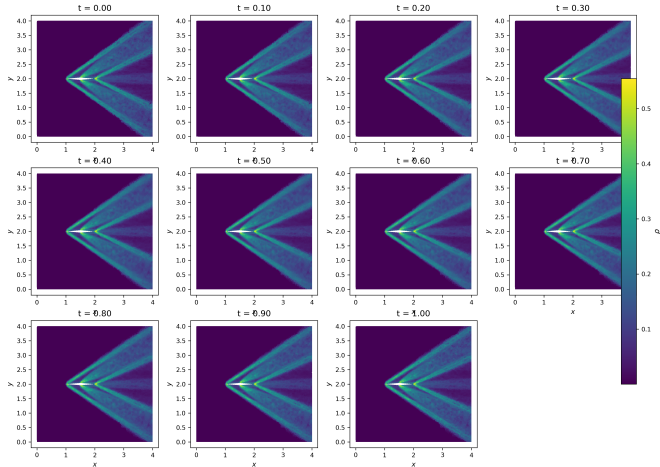


Absolute Error — $M : 2.90 \rightarrow 3.00$ (high)

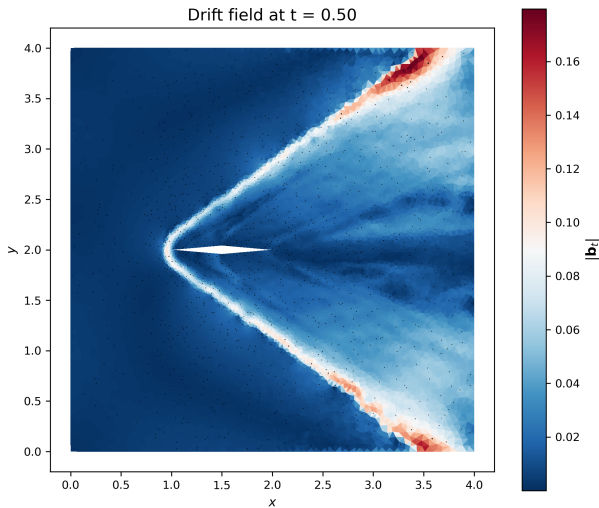


Density Transport — $M : 2.00 \rightarrow 2.10$

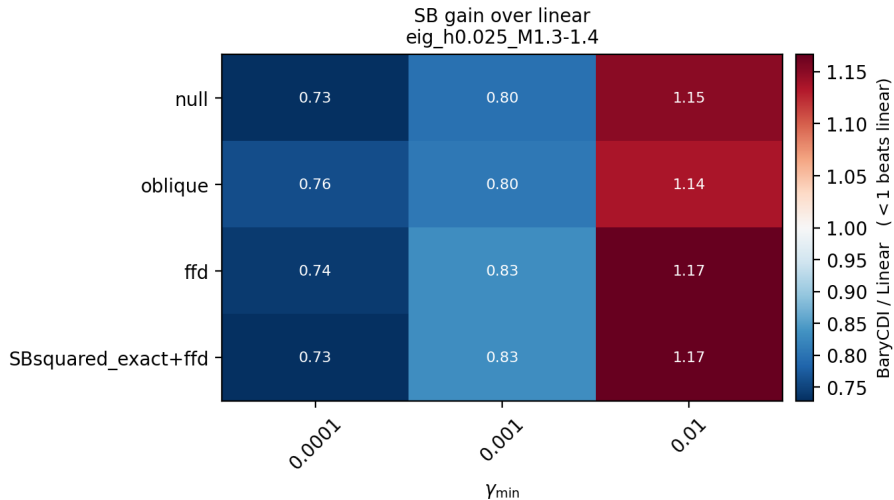
2D density transport



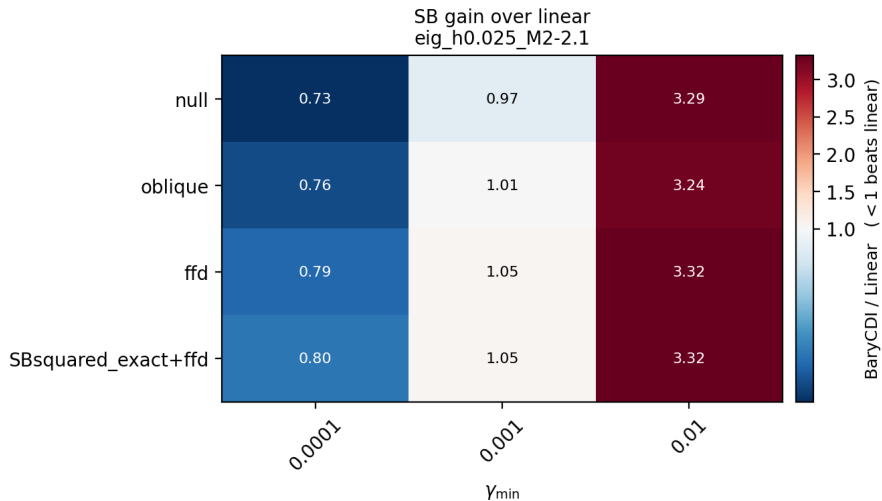
Reference Drift Field — $M : 2.00 \rightarrow 2.10$



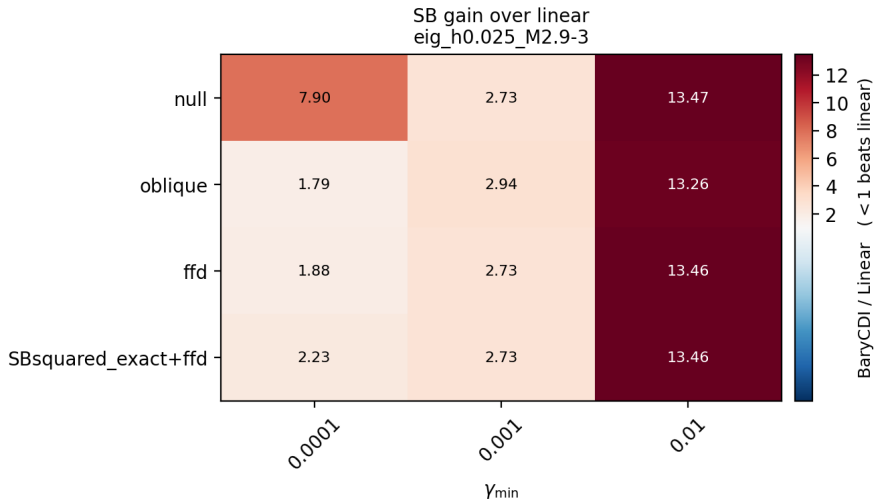
Gain over Linear — $M : 1.30 \rightarrow 1.40$



Gain over Linear — $M : 2.00 \rightarrow 2.10$



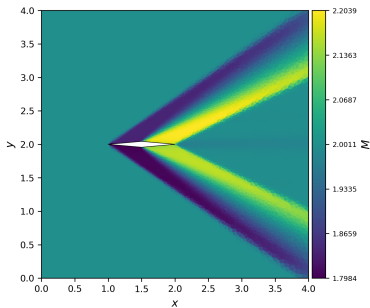
Gain over Linear — $M : 2.90 \rightarrow 3.00$



Snapshots — AoA $0^\circ \rightarrow 1^\circ$, $t = 0.5$

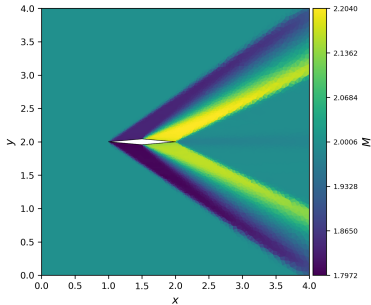
BaryCDI

SB-CDI (barycentric) $t=0.50$



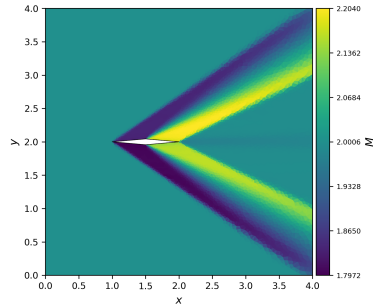
FFD

FFD registration $t=0.50$

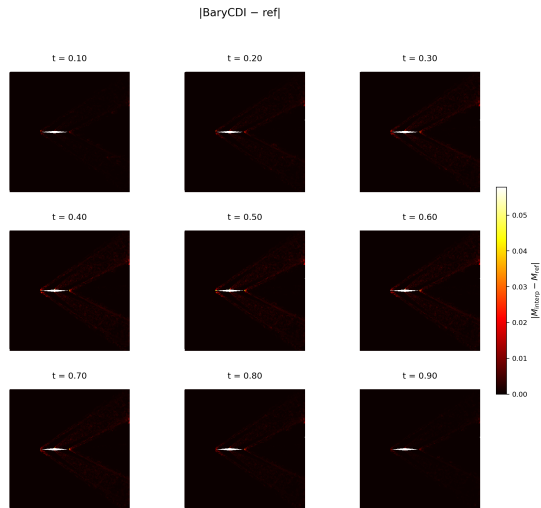


Linear

Linear $t=0.50$



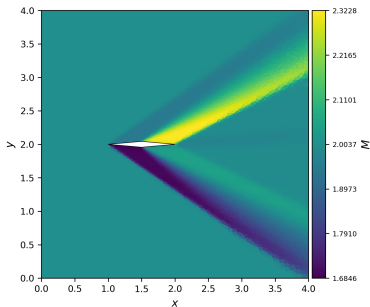
Absolute Error — AoA $0^\circ \rightarrow 1^\circ$



Snapshots — AoA $3^\circ \rightarrow 4^\circ$, $t = 0.5$

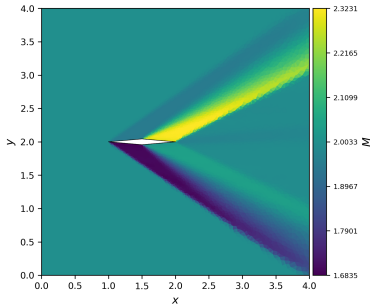
BaryCDI

SB-CDI (barycentric) $t=0.50$



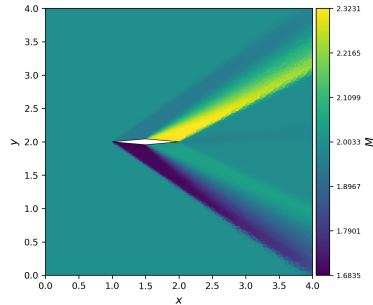
FFD

FFD registration $t=0.50$

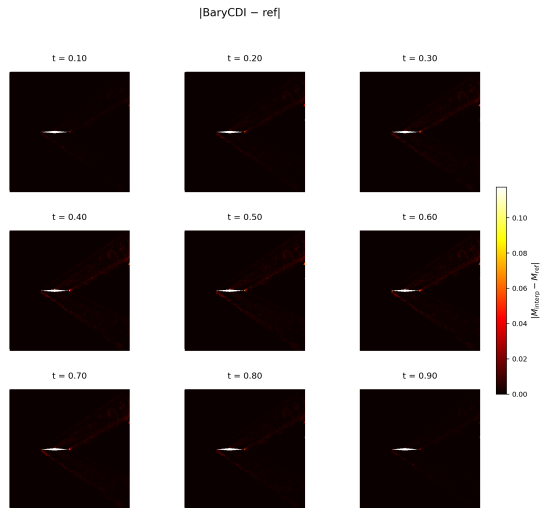


Linear

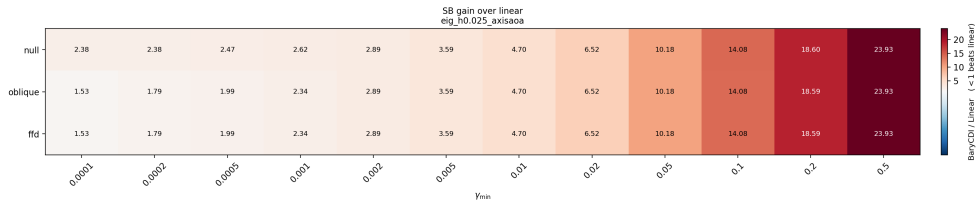
Linear $t=0.50$



Absolute Error — AoA $3^\circ \rightarrow 4^\circ$

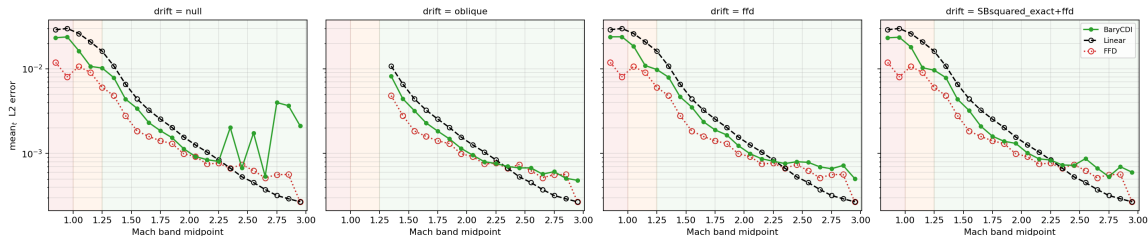


Gain over Linear — AoA Axis



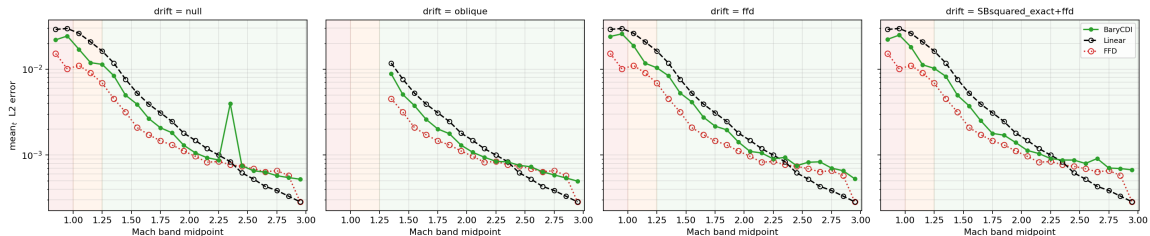
Error vs. Mach Interval (AoA 0° , $\gamma_{\min} = 10^{-4}$)

L2 error vs Mach interval ($\gamma_{\min} \approx 0.0001$, AoA 0°)
red: no shock ($M < 1$) orange: detached bow shock green: attached oblique shock (validated regime)

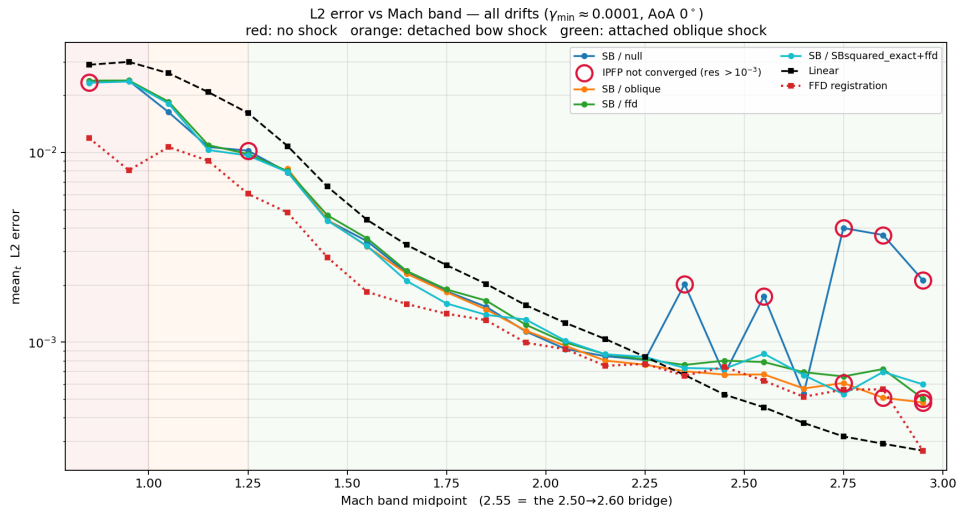


Error vs. Mach Interval ($\text{AoA } 2^\circ$, $\gamma_{\min} = 10^{-4}$)

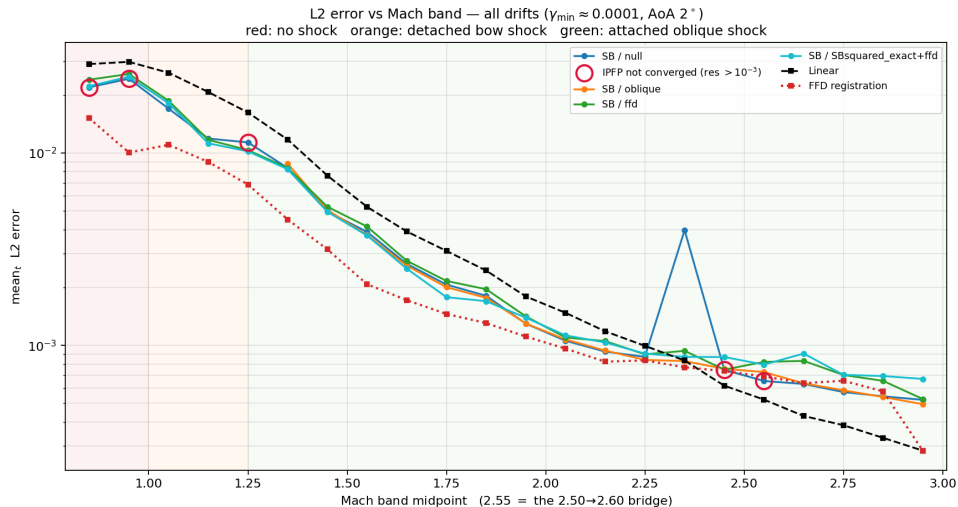
L2 error vs Mach interval ($\gamma_{\min} \approx 0.0001$, $\text{AoA } 2^\circ$)
red: no shock ($M < 1$) orange: detached bow shock green: attached oblique shock (validated regime)



Error vs. Mach — All Drifts on One Axis



Error vs. Mach — All Drifts, AoA 2°

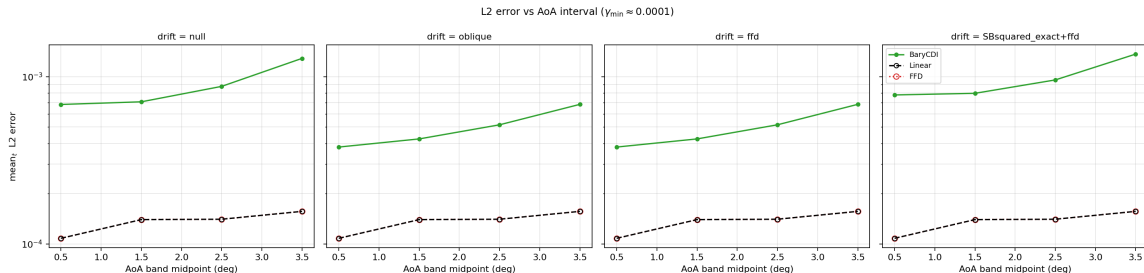


Error vs. Mach — the Crossover

band	BaryCDI	Linear	FFD	best
0.80–0.90	2.39×10^{-2}	2.89×10^{-2}	1.19×10^{-2}	FFD
1.50–1.60	3.52×10^{-3}	4.41×10^{-3}	1.84×10^{-3}	FFD
2.20–2.30	8.11×10^{-4}	8.39×10^{-4}	7.69×10^{-4}	FFD
2.40–2.50	8.00×10^{-4}	5.31×10^{-4}	7.41×10^{-4}	Linear
2.90–3.00	5.05×10^{-4}	2.69×10^{-4}	2.69×10^{-4}	Linear

Error falls ~ 2 decades from $M \approx 0.85$ to $M \approx 2.95$. Above $M \approx 2.45$ **Linear wins**: within a 0.10 band at high Mach the shock barely moves (β_{LE} : $34.3^\circ \rightarrow 23.1^\circ$ over $M=2 \rightarrow 3$), so transport can only add error.

Error vs. AoA Interval ($\gamma_{\min} = 10^{-4}$)



Which Drift? — Paired per-Band Comparison

BaryCDI mean- t L2, paired by band, ties excluded, sign test.

$\gamma_{\min}$	comparison	wins	median ratio	p	verdict
10^{-4}	oblique vs ffd	33/39	0.931	$< 10^{-4}$	oblique +6.9%
10^{-4}	oblique vs SB^2 +ffd	26/38	0.930	0.034	oblique +7.0%
10^{-4}	ffd vs SB^2 +ffd	16/48	1.021	0.029	SB^2 +2.1%
10^{-3}	oblique vs ffd	34/38	0.947	$< 10^{-4}$	oblique +5.3%
10^{-3}	ffd vs SB^2 +ffd	<i>identical</i> — seed fixed above γ_*			
10^{-2}	oblique vs ffd	33/34	0.969	$< 10^{-4}$	oblique +3.1%

oblique wins at every γ

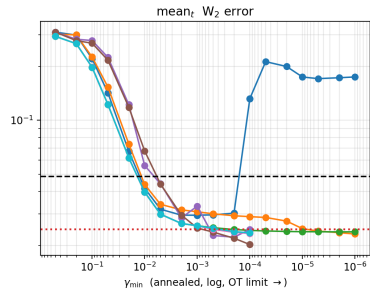
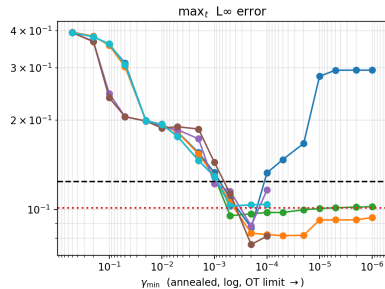
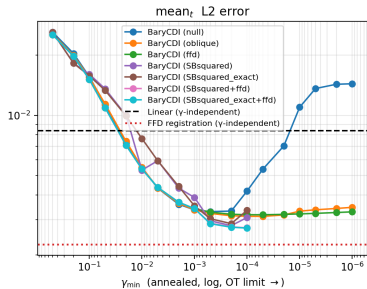
Margin grows as $\gamma \downarrow$: $3.1 \rightarrow 5.3 \rightarrow 6.9\%$.
Only defined for $M \gtrsim 1.25$.

SB^2 +ffd $\approx$ ffd

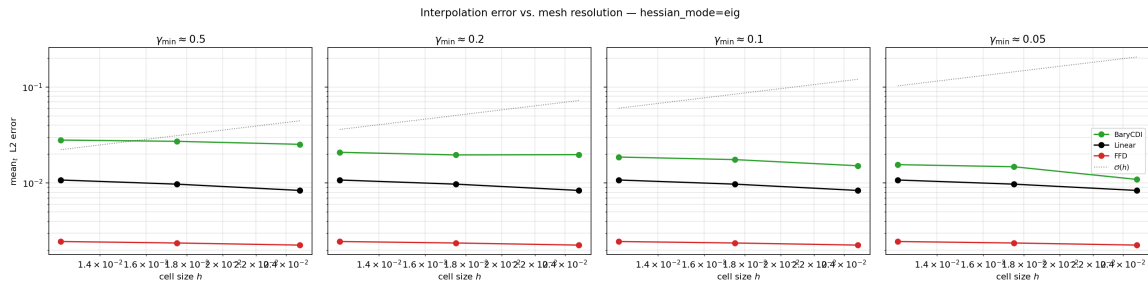
+2.9% on the Mach axis, but -93% on the AoA axis (median ratio 0.517), for a full extra anneal.

Error vs. $\gamma_{\min}$

Interpolation error vs. annealed $\gamma_{\min}$ — hessian_mode = eig_h0.025

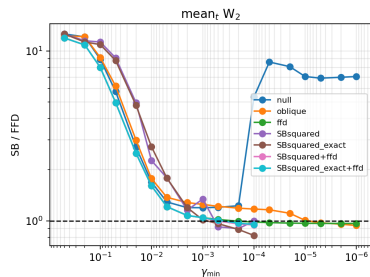
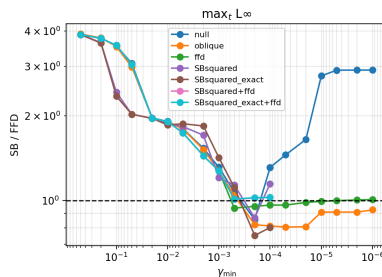
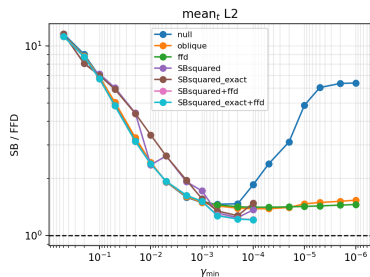


Error vs. Mesh Resolution h

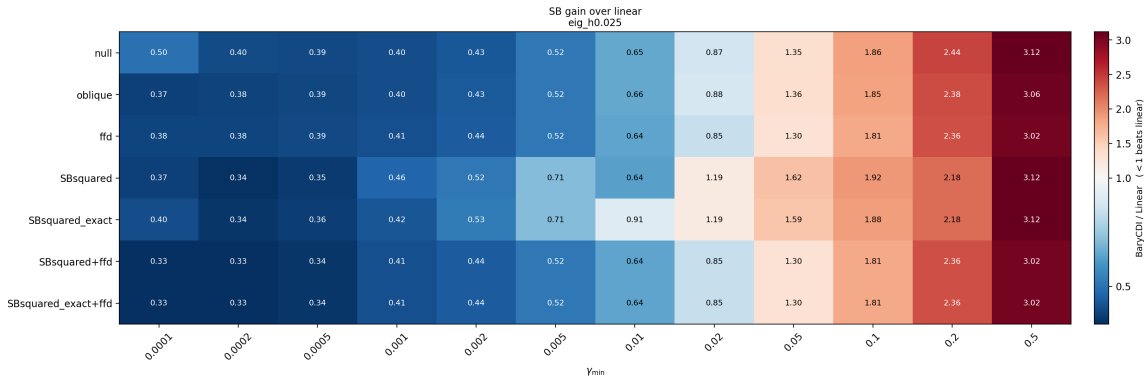


SB vs. FFD Registration

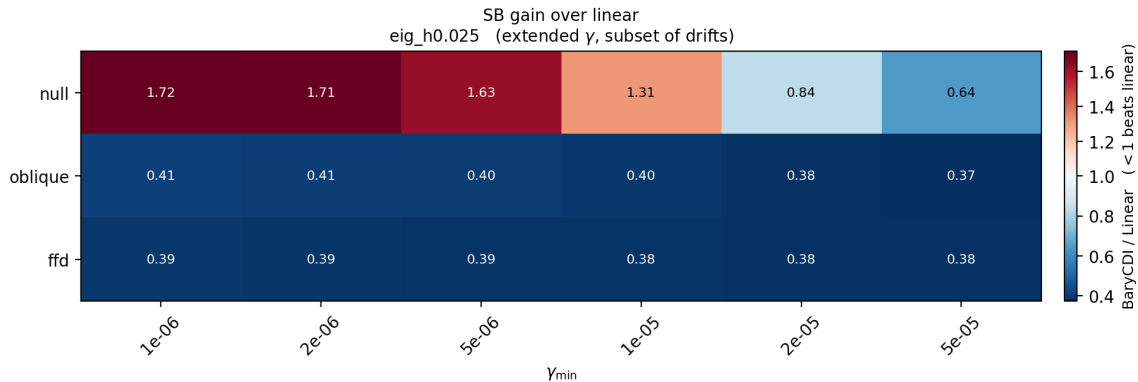
Schrödinger bridge vs. raw FFD registration (<1 = SB better) — hessian_mode = eig_h0.025



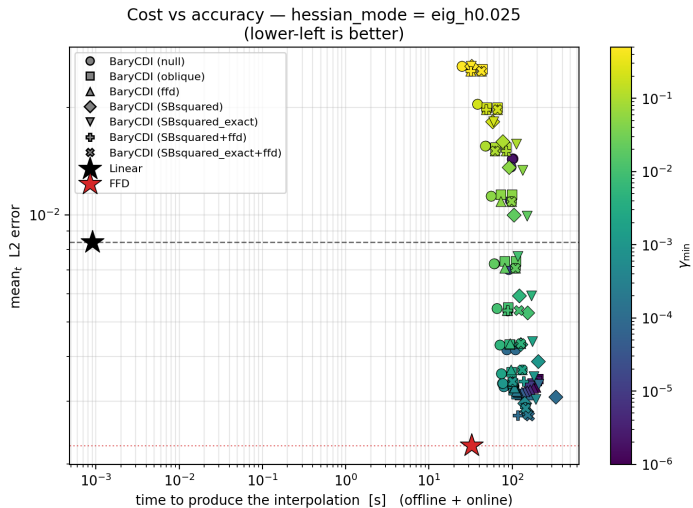
Gain over Linear — Heatmap



Gain over Linear — Extended γ (null / oblique / ffd only)

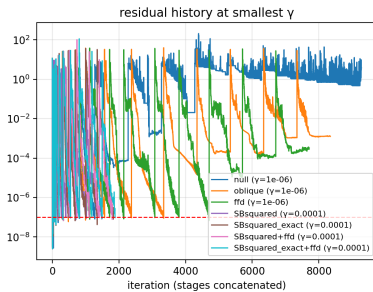
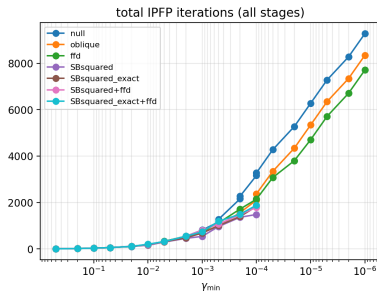
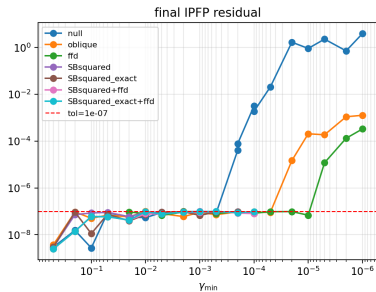


Cost vs. Accuracy



IPFP Convergence

IPFP convergence — hessian_mode = eig_h0.025



- **Regime dominates.** Error falls ~ 2 decades from $M \approx 0.85$ to 2.95 ; sub- $M1$ bands are $30-100\times$ worse and lie outside the attached-shock premise the marginal is built on.
- **Linear wins above $M \approx 2.45$** — transport buys nothing once the shock barely moves across the band.
- **On the AoA axis SB loses to Linear by $4-8\times$** , and the FFD registration recovers ≈ 0 displacement (FFD $\equiv$ Linear).
- **oblique is the best drift** at every γ ($+3-7\%$, $p < 10^{-4}$); null is $+26\%$ worse and unstable at high Mach.
- **The raw FFD registration still beats the bridge built on it** (1.24×10^{-3} vs 1.72×10^{-3} on matched bands).